

CSC3066 Deep Learning

Autoencoders

Lu Bai (l.bai@qub.ac.uk)

Content of Today Lecture

- ❑ Unsupervised Learning & Dimensionality Reduction
- ❑ Introduction to Autoencoders
- ❑ Applications of Autoencoders



Supervise vs Unsupervised Learning

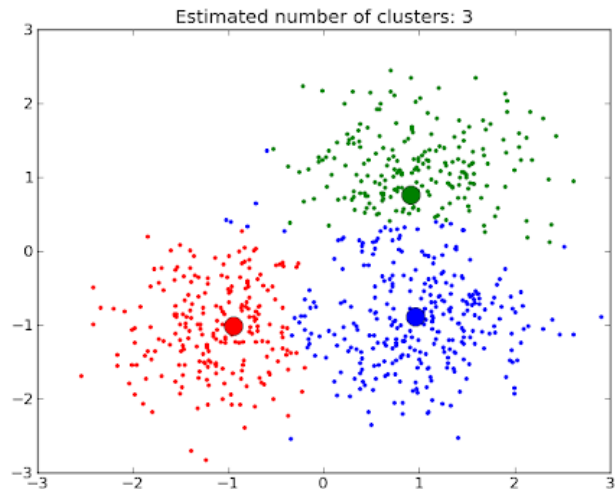
- ❑ Unsupervised Learning
 - ❑ Data: x , no labels
 - ❑ Goal: Learn some underlying hidden structure of the data
 - ❑ Examples: Clustering, dimensionality reduction, feature learning, etc.



Supervise vs Unsupervised Learning

- Unsupervised Learning

- Clustering example:

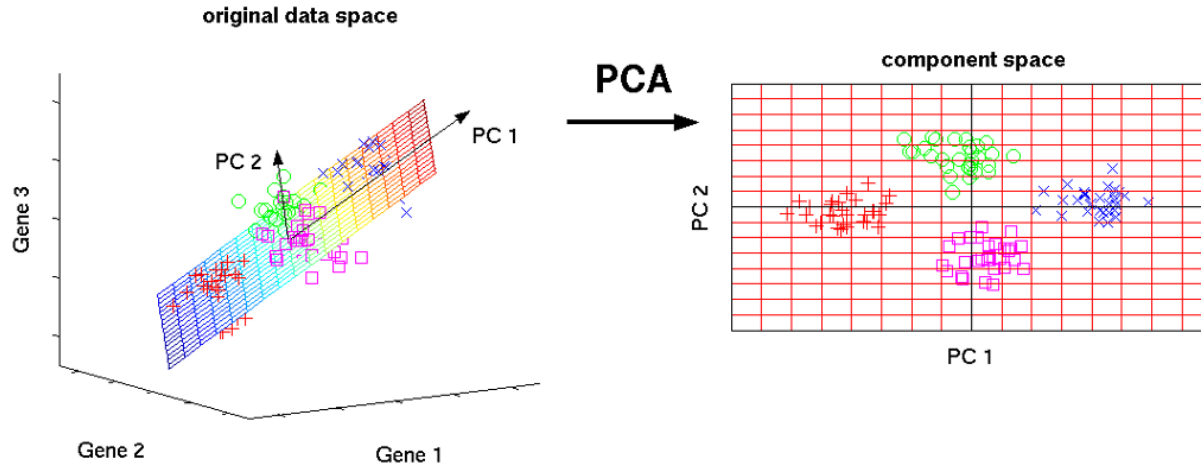




Supervise vs Unsupervised Learning

□ Unsupervised Learning

□ Dimensionality Reduction:





Supervise vs Unsupervised Learning

☐ Unsupervised Learning

- ☐ Why dimensionality reduction?

- ☐ Many modern data domains involve huge numbers of features / dimensions:

- Documents: thousands of words, millions of bigrams
- Images: thousands to millions of pixels
- Genomics: thousands of genes, millions of DNA polymorphisms



Supervise vs Unsupervised Learning

☐ Unsupervised Learning

☐ Why dimensionality reduction?

☐ High dimensionality has many costs

- Redundant and irrelevant features degrade performance of some ML algorithms (higher risk of overfitting)
- Difficulty in interpretation and visualization
- Computation may become infeasible
- It takes much larger space to store data with a large number of features



Definition of Autoencoders

- ❑ Autoencoder:
 - ❑ Autoencoder is an Artificial Neural Network capable of learning efficient (reduced) representation of the input data
 - ❑ It is also capable of generating new data that looks very similar to the train data, i.e. data from the same distribution (*generative model*).



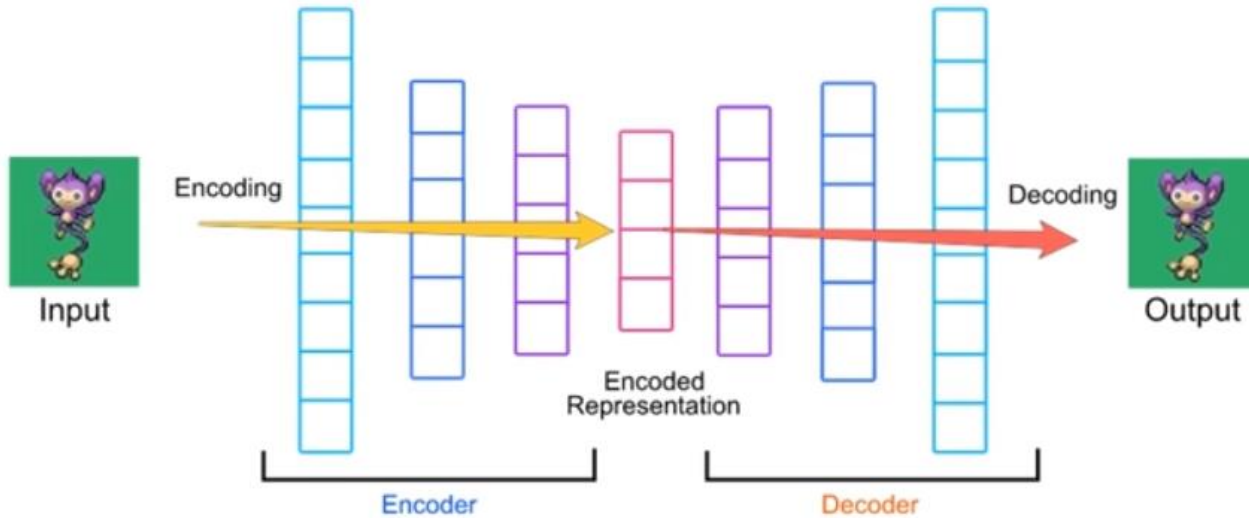
Definition of Autoencoders

- ❑ Autoencoder:
 - ❑ Autoencoder is a network trained to attempt to copy its input to its output.
 - ❑ It is an unsupervised model (doesn't need any labels), however, it is trained using supervised learning method.
 - ❑ It applies backpropagation, setting the output value to be equal to the input.



Definition of Autoencoders

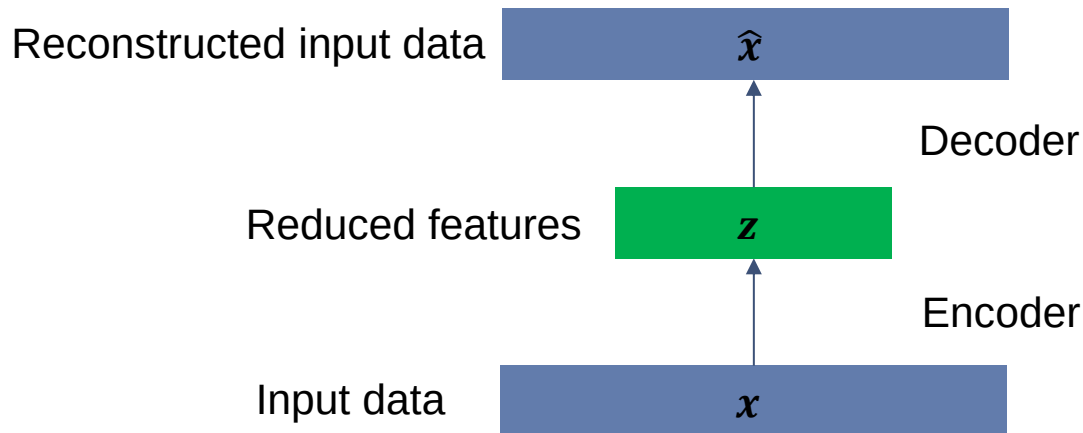
□ Autoencoder:





Definition of Autoencoders

- Autoencoder: General Idea
 - Train a model so that the new (compressed) feature representation can be used to reconstruct our original data





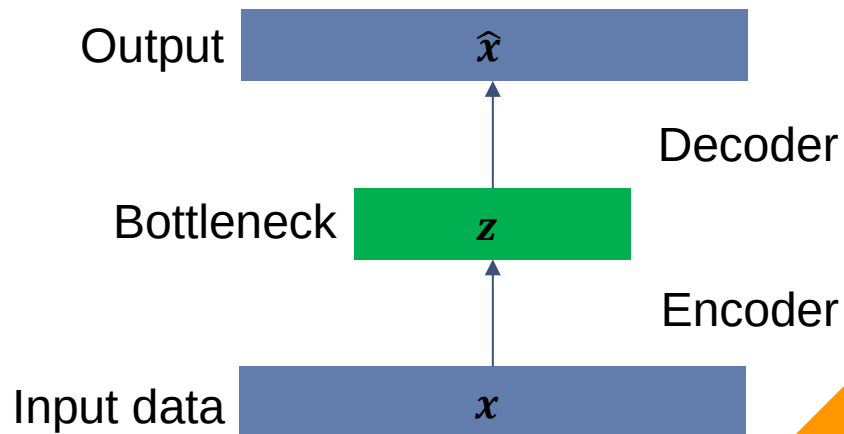
Architecture of Autoencoder

- Components of Autoencoder:

- Encoder

- Code/bottleneck

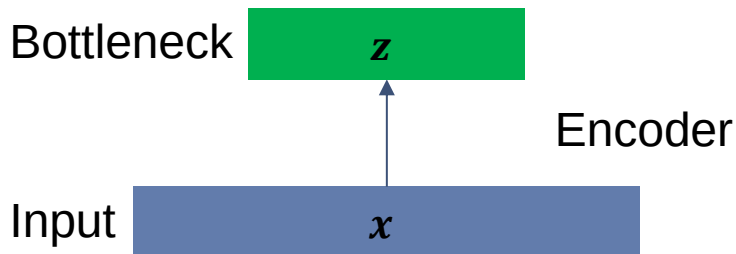
- Decoder





Architecture of Autoencoder

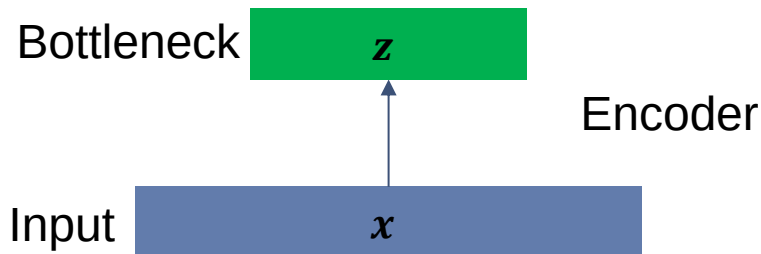
- ❑ Autoencoder: Encoder
 - ❑ Encoder is a network that maps the input into a lower-dimensional feature representation z
 - ❑ The space represented by these fewer features is often called the latent-space, bottleneck or codings.





Architecture of Autoencoder

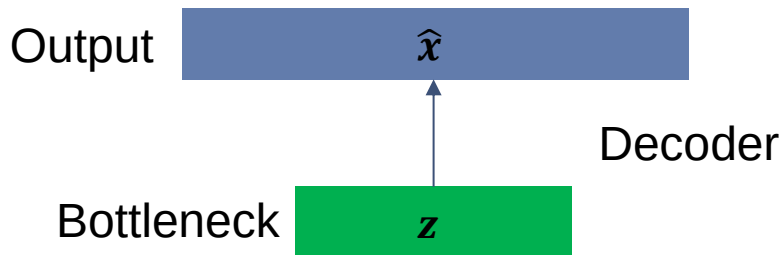
- ❑ Autoencoder: Code
 - ❑ Represents the latent-space
 - ❑ Learned compressed features z (usually smaller than x)
 - ❑ z should represent the most important features from x





Architecture of Autoencoder

- ❑ Autoencoder: Decoder
 - ❑ Decoder is a network that tries to reconstruct the input using only the encoding of the input.
 - ❑ Goal: for an input z output $\hat{x}$ of the same dimension as x and as similar as possible to x .





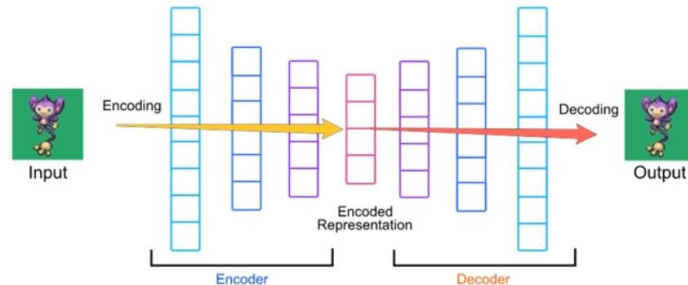
Architecture of Autoencoder

- ❑ Autoencoder:
 - ❑ Encoder and Decoder can be any type of Neural Network (MLP, CNN, RNN)
 - ❑ They can have multiple hidden layers (stacked autoencoders)
 - ❑ Adding more layer helps autoencoder learn more complex codings, however, it increases the risk of overfitting
 - ❑ Encoder and Decoder are typically symmetrical but it is not necessary



Architecture of Autoencoder

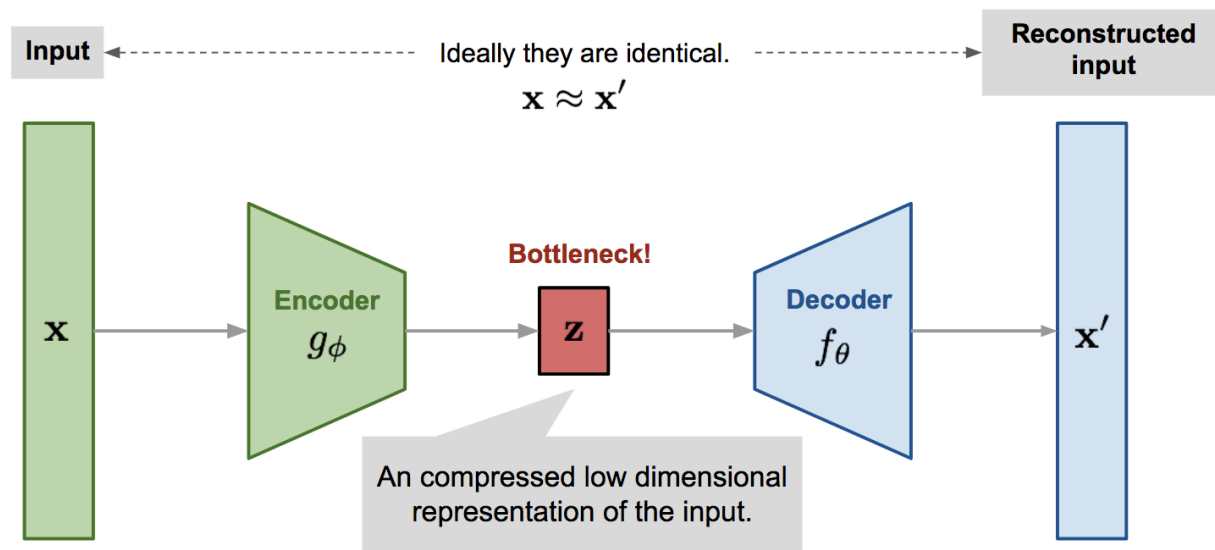
- ❑ Autoencoder:
 - ❑ The dimensions of the input layer and the output layer are the same
 - ❑ Hidden layers have dimension smaller than the dimension of the input





Architecture of Autoencoder

□ Autoencoder:





Training Autoencoder

□ Autoencoder:

Mathematically the encoder and the decoder can be defined as transitions g and f , such that:

$$g: X \rightarrow F$$

$$f: F \rightarrow X$$

$$g, f = \underset{g, f}{\operatorname{argmin}} \|X - (f \circ g)X\|^2$$

X – original space, F – latent space, $(f \circ g)X \rightarrow f(g(X))$



Training Autoencoder

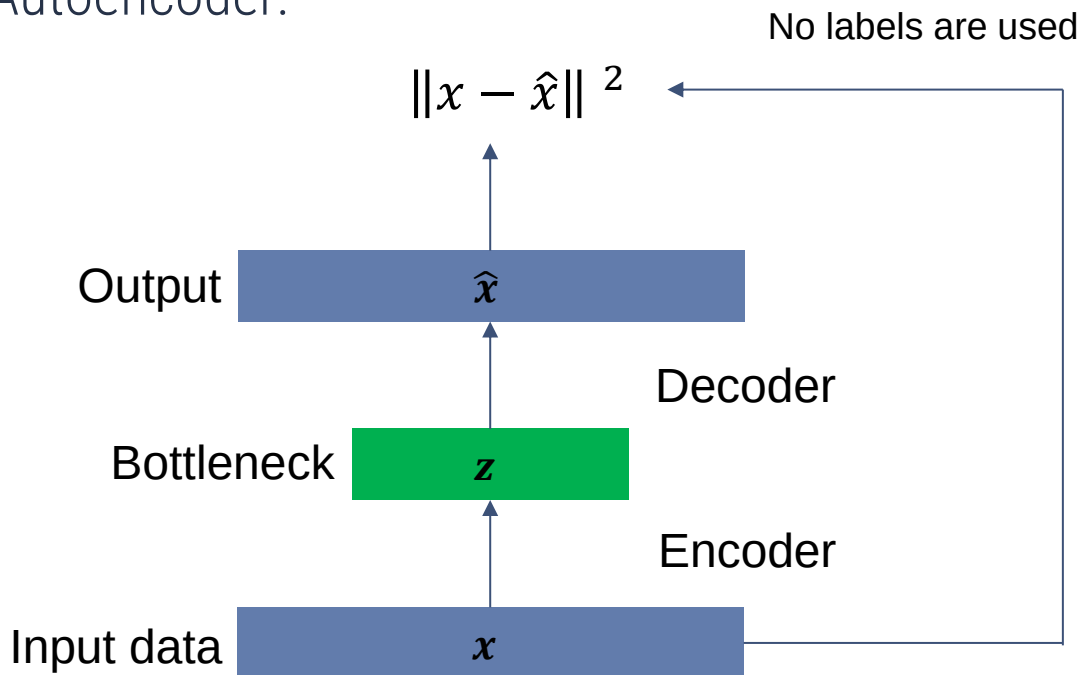
- ❑ Autoencoder:
 - ❑ The aim of the autoencoder is to select the encoder and decoder functions in such way that $f(g(x))$ is as similar to x as possible.
 - ❑ The learning process of autoencoder is the process of minimizing the mean squared loss function:

$$\|X - (f \circ g)X\|^2$$



Training Autoencoder

□ Autoencoder:





Training Autoencoder

- ☐ Parameters of Autoencoders:
 - ☐ Number of layers
 - ☐ Number of nodes per layer
 - ☐ Activation function
 - ☐ Loss function (mean squared error)
 - ☐ Bottleneck size



Application of Autoencoders

- ☐ Dimensionality Reduction
- ☐ If the training is successful, the autoencoder has learned how to represent the input values in a different, yet more compact form.
- ☐ This compact representation could be then used for further analysis, e.g. can be applied with any ML models.



Application of Autoencoders

- ❑ Data Denoising (commonly applied with images)
 - ❑ Instead of minimizing $L(X - (f \circ g)X)$, where L is a loss function penalizing $(f \circ g)X$ from being dissimilar from X , denoising autoencoder minimizes:

$$L(X - (f \circ g)\tilde{X})$$

where $\tilde{X}$ is a copy of X which was corrupted with some form of noise.



Application of Autoencoders

- ❑ Image (data) Generation
 - ❑ Variational Autoencoder (generative model):
autoencoder whose training is regularised to avoid overfitting and ensure that the latent space has good properties that enable generative process
 - ❑ E.g. given input images (e.g., images of face or scenery), the system will generate similar images

Coming Next

This was our last lecture!